Physics: Newton’s Laws of Motion - Page 1

Classical Mechanics: Principles of Motion and Dynamics
Sir Isaac Newton published his three fundamental laws of motion in 1687.
These laws govern the motion of all macroscopic objects.

First Law (Law of Inertia):
An object remains at rest or in uniform straight-line motion unless acted
upon by a non-zero net external force. Mass is the measure of inertia.

Second Law (Force and Acceleration):
The net force applied to an object equals its mass multiplied by its acceleration:
F = m * a
Force is measured in Newtons (N), where 1 Newton equals 1 kg * m/s^2.

Physics: Newton’s Laws of Motion - Page 2

Third Law (Action and Reaction):
For every action force, there is an equal and opposite reaction force.
Whenever body A exerts a force on body B, body B exerts an equal force in the opposite direction on body A.

Important rule: Action and reaction forces act on different bodies, which is why they never cancel each other out.
Example: A rocket accelerates forward because its engines expel exhaust gases backward.